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Review Article

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The Origin and Early Evolution of Life: Homochirality Emergence in Prebiotic Environments

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Abstract

Homochirality is one of the signatures of life. Numerous geological and prebiotic chemistry studies have proved that disordered soups containing small organic molecules, gases, liquids, and minerals (such as those containing phosphorous) yielded racemic mixtures of building blocks for biomolecule assembly. Polymers obtained from these bricks should have been enantiopure with functional properties similar to modern biomolecules or heterochiral with some functions such as catalyzing a chemical transformation unspecifically. Up until now, no clues have been found as to how symmetry breaking occurred. In this review, we highlight the principal achievements regarding the emergence of homochirality during the prebiotic synthesis of building blocks. Furthermore, we tried to focus on approaches based on prebiotic systems chemistry (bottom-up) and laboratory scales to simulate plausible prebiotic messy environments for the emergence of life. We aim with this review to assemble, even partially, the puzzle pieces of the origin of life regarding the relevant phenomenon of homochiral symmetry breaking. Key Words: Homochirality—Symmetry breaking—Enantiomers—Amino acids—Phospholipids. Astrobiology 23, 1368–1382.

1. Introduction

In modern cells, nucleic acids, lipids (and thus membranes), peptides, and proteins are homochiral. On the contrary, prebiotic environments can be considered heterogeneous ''soups'' displaying mixtures of unselected and achiral compounds. Thus, symmetry breaking occurred at some point during the biological chemical evolution of some biomolecules, such as the aforementioned polymers (DNA, RNA, and peptides). Further, since homochirality was selected through ''chemical'' evolution before the emergence of the first forms of life, it seems fair to assume that heterochiral and homochiral molecules have distinct properties, and that homochirality was more suited. Of note, it cannot be neglected that for another form of life on an exoplanet, D-amino acids (for instance) could have been selected.

In this review, we discuss the unnatural, yet necessary, link between these heterogeneous environments and homochirality's emergence. The spontaneous generation of homochirality in an achiral environment is not an intuitive concept. It seems that in the absence of any chiral-inducing agent, symmetry breaking is not possible to occur. It can be argued that spontaneous homochirality emergence happened in a heterogeneous environment such as space interstellar clouds, deep-sea smokers, or hydrothermal pools where organic (carbon-based compounds) and inorganic (essentially minerals) compounds could have interacted due to external forces such as radiation, heat, or both (Lopez and Fiore, 2019; Ozturk and Sasselov, 2022).

For a chemist, symmetry breaking is the spontaneous generation of an excess of one enantiomer over the other. Considering biomolecules, α-amino acids proved to be a fertile class of biomolecules to investigate homochirality emergence, along with sugars (Breslow and Cheng, 2010; Hein and Blackmond, 2012; Meinert *et al.*, 2016; Kawai, 2017; Jiménez-Serra *et al.*, 2020; Martínez *et al.*, 2022).

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More recently, phospholipids and phospholipid precursors were added to the list (Fiore and Buchet, 2020). Thus far, there would be two possible explanations for the origin of chiral symmetry breaking in the cosmos: (i) the parity-violating energy difference between more stable molecules (*i.e.*, L-amino acids are more stable than D-amino acids), based on parity violation in the weak interactions; and (ii) the degradation of one enantiomer caused by chiral cosmic forces such as circularly polarized light (CPL), resulting in enantiomeric enrichment (0.1% ee) [$ee = (C_D - C_L)/(C_D + C_L)$] with C_D and C_L as the concentration of both enantiomers of (again) the L- over D-amino acids. According to Bonner (1994), chirality can be induced in organic molecules by photosynthesis or photolysis using left or right CPL.

To validate these theories, reliable and reproducible significant enantiomeric amplification mechanism models—not yet developed—must be established. In addition, an aspect that cannot be neglected is that the amplification of a slight enantiomeric imbalance in enantiomers is based on asymmetric autocatalysis.

We have noticed that the recent literature lacks a summary of relevant experiments carried out in plausible prebiotic conditions that suggest how the phenomenon of homochiral symmetry breaking can be studied (Martínez *et al*., 2022) according to the process of prebiotic evolution of biomolecules (Ruiz-Mirazo and Moreno, 2012). Under young Earth conditions, the atmospheric composition and the meteorite bombardment (4.1–3.8 Ga ago) could be considered as chemical and energy sources at the same time. These conditions may have facilitated the production of nitrogen-reduced compounds, such as HCN and cyanoacetylene. Such products could accumulate and concentrate in lakes or enclosed pools (Spitzer, 2011; Westall *et al*., 2018). The prebiotic conditions needed for the synthesis of those chiral molecules that are today considered necessary bricks of life have always been under debate.

The first example to experimentally reproduce the abiotic synthesis of biomolecules is the famous Miller (and Urey) experiment, which consisted of a strongly reduced mixture of gases (methane, ammonia, and hydrogen) (Airapetian *et al*., 2016) stimulated by electrical sparks to form amino acids (Fig. 3B) (Miller, 1953). The idea was to simulate an environment that could have been present on early Earth (Hadean or early Archaean). Meanwhile, pathways to produce purines as nucleobases for nucleic acids were presented by Oró and Orgel (Fig. 1A) (Oró, 1960; Orgel *et al*., 1967) and based on the polymerization of HCN (**1**, Fig. 1A). Pyrimidines, instead, were obtained from cyanoacetylene (Larralde *et al*., 1995; Robertson and Miller, 1995). Sugars, among them ribose, can be products of the formose reaction of formaldehyde (**2**, Fig. 1C) (Breslow, 1959).

The three pathways to produce sugars, amino acids, and nucleobases have been previously described as the pillars of prebiotic chemistry (Islam and Powner, 2017) as they represent ways of synthesizing, abiotically, the main components of RNA and peptides. Even if they are extremely important in this matter, the three pathways still suffer from some downsides. First, in all three routes, the biomolecules of interest are detected but are rather unstable, so they continue evolving, forming intractable and incontrollable mixtures (Fig. 1A–C). Second, as in the case of Miller's experiment, the concentrations required for the reactants are much higher than those realistically achievable in primitive environments. Furthermore, amino acids can easily form glyconitrile (**3**, Fig. 1B), which is an intermediate for polymerization in Strecker's type reaction (*vide infra*).

2. L for Amino Acids, D for Sugars, and 2*R* and 2*S* for Phospholipids: Introducing the Chiral Actors

Chiral molecules in living organisms exist almost exclusively as a single set of enantiomers, a critical property for

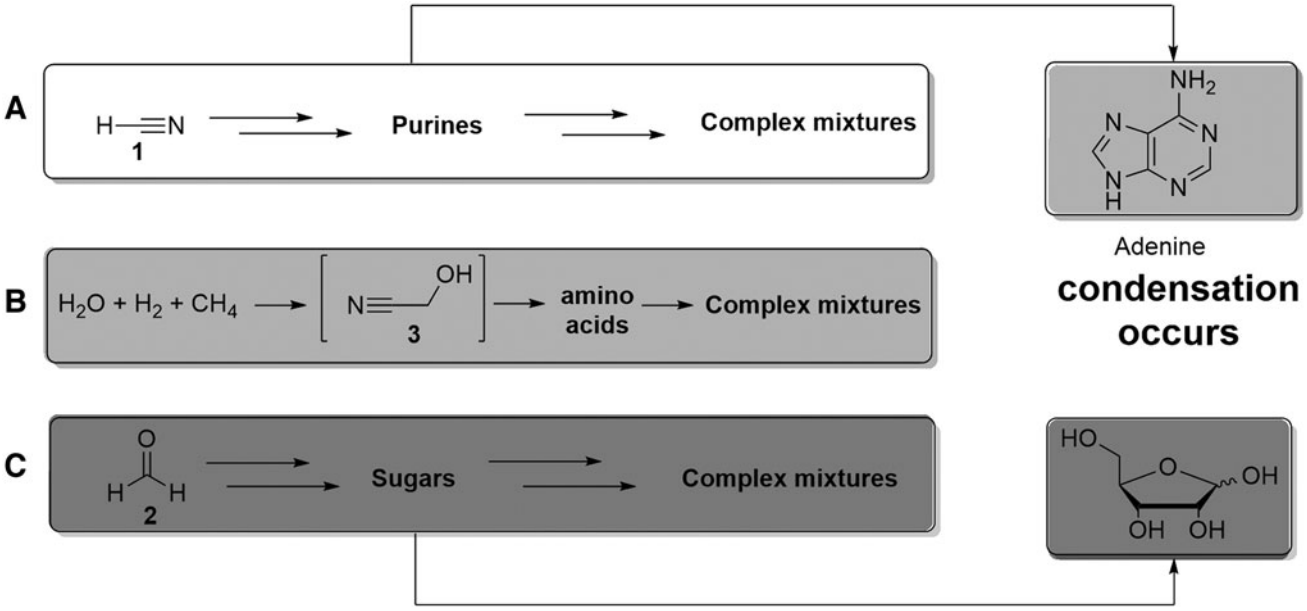

FIG. 1. Three pillars of prebiotic chemistry: **(A)** purines through Oro' and Orgel's synthesis from cyanide (**1**) (Oró, 1960); **(B)** amino acids synthesis through Miller's experiment (Miller, 1953); **(C)** sugar formation through ''formose'' polymerization of formaldehyde (**2**) (Breslow, 1959).

molecular recognition and replication processes (Mason, 1982; Janoschek, 1991; Bonner, 1994; Avetisov and Goldanskii, 1996). Sugars, such as ribose or 2-deoxy ribose, possessing the D configuration only, constitute the chiral part of nucleotides such as adenosine triphosphate (Fig. 2A) (Altamura *et al*., 2021), and are the chiral part of RNA and DNA. Otherwise, amino acids are the constituents of peptides, such as glutathione (Fig. 2B) (Fiore *et al*., 2011), and proteins. They are present in proteins only in the L configuration. Recently, the attention to symmetry breaking of chiral molecules was drawn to phospholipids. Phospholipids contain, as well as α-amino acids, one single chiral carbon: the middle atom of the glycerol backbone (excluding the chiral lipidic chains in some archaeal lipids, Fig. 2C, D) (Fiore *et al*., 2022).

The most common natural enantiomer for glycerol-based phospholipids remains D (Fischer's convention, based on the handedness of glyceraldehyde as for sugars), and is defined as 1,2-diacyl-*sn*-glycerol-3-phosphoalcohol (in *sn*-glycerol nomenclature) (IUPAC-IUB Commission on Biochemical Nomenclature, 1968) or 2*R* (Kennedy, 1956; Weis and McConnell, 1984) in the Cahn–Ingold–Prelog formalism, Fig. 2C. The opposite configuration of L-dialkylglycerol is defined as a 2,3-dialkyl-*sn*-glycerol-phosphoalcohol or 2*S*. This symmetry occurs for the natural lipidic isoprenoid glycerol ethers of Archaea membranes (Fig. 2D) (Lombard *et al*., 2012; Fiore and Buchet, 2020). The relevance of enantiopure membranes and the relevance of their chemical and physical aspects were recently summarized and will not be further discussed in this section (Martin *et al*., 2021).

3. Autocatalysis: From a Mathematical Model to Autocatalytic Reactions

During the early 1950s, Frank (1953) suggested the first mathematical model for chiral autocatalysis. He supposed that each enantiomer catalyzes its formation suppressing the formation of the opposite enantiomer: for example, *R* versus *S*, as shown in Fig. 3 (Babinec and Krempaský, 1994). More than 20 years later, Soai set up a series of autocatalytic systems where a catalytic amount of enantiopure aromatic dialkyl zincs was added to aromatic aldehydes. The most intriguing fact is that the experimental work of Soai is, so far, the only autocatalytic reaction that supports the mathematical model of Frank (Soai *et al*., 2019). In brief, with the reaction that was named after his inventor, a chiral alcohol can be produced with a large ee% using the propylzincate enantiopure intermediate as a catalyzer. The first trials gave modest results (35% ee after subtracting the contribution of the catalyst). Later on, the system was set up to implement the fee from 51% to 99.5% using not only a catalytic amount of the enantiopure diisopropyl zincate (Fig. 4) but also pure, nonaromatic, and nonpolar compounds such as L-valine and L-leucine. Cumene was also used instead of toluene (Fig. 4) as a solvent. Soai himself (Soai *et al*., 2019), or together with Gridnev *et al.* (2004) and Blackmond (2020) and later Denmark (Athavale *et al.*, 2020), tried to rationalize the tetrameric model proposed to explain this autocatalytic reaction. Following the kinetic studies of the Soai reaction, other kinetic models have been suggested for the asymmetric amplification of enantiomeric excess in an autocatalytic process.

The first comprehensive mechanistic study of the autocatalytic behavior of the Soai reaction came from Blackmond and Brown (Blackmond *et al*., 2001; Gridnev *et al*., 2004; Blackmond, 2006, 2020). The main features of the proposed reaction system are as follows: (i) the alkanol product of the reaction is driven to form dimers or tetramers; (ii) for pyramidal substrate $R = CH_3$, heterochiral and homochiral dimers and tetramers are formed stochastically, that is, resulting in an equal preference for *R* and *S*; (iii) homochiral dimers and tetramers are active in autocatalytic reactions in

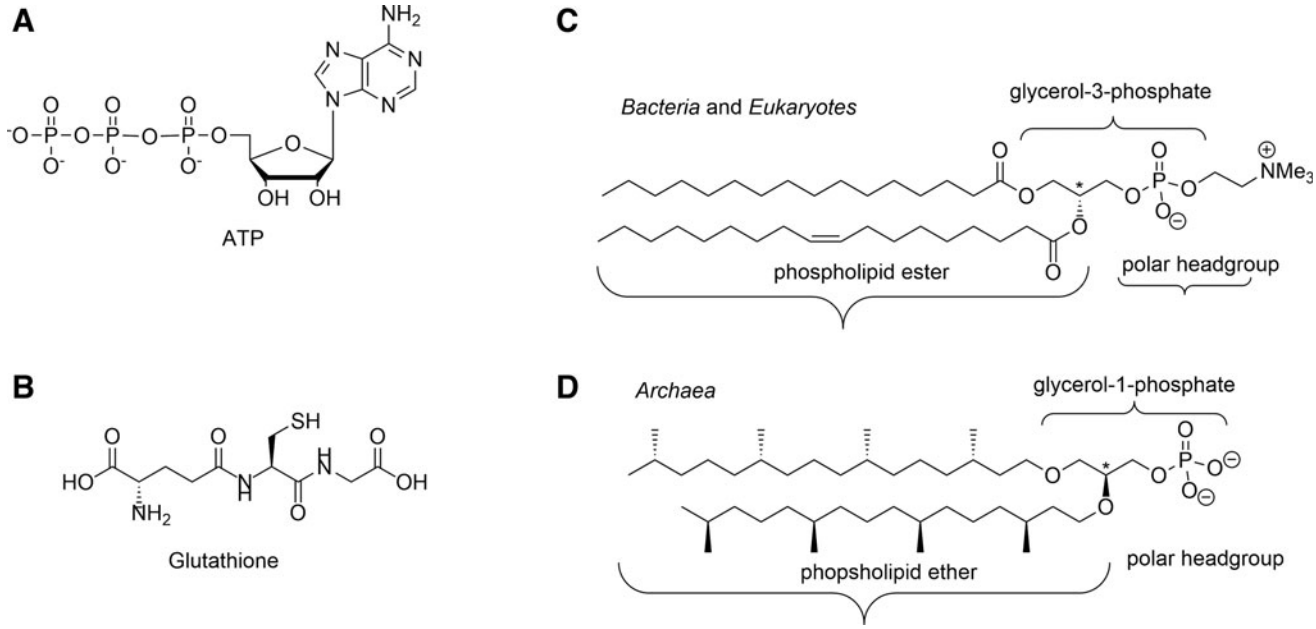

FIG. 2. Chemical structures and absolute configurations of D-ribose in ATP **(A)**; L-amino acids in glutathione **(B)**; D-glycerol in a phospholipid ester **(C)** and L-glycerol in a phospholipid ether **(D)**. ATP = adenosine triphosphate.

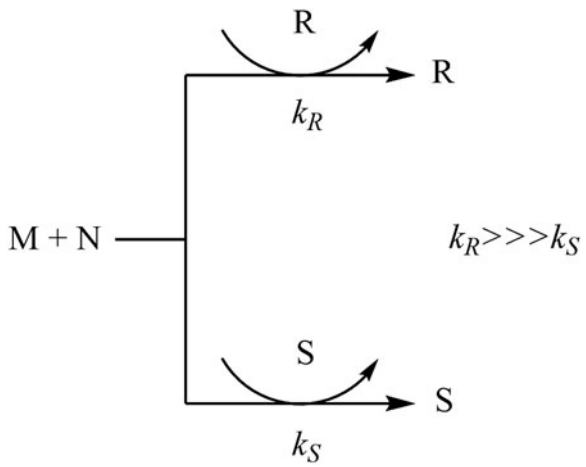

FIG. 3. Frank's autocatalytic model; redraw as reported in Frank (1953).

which the enantiomeric excess is amplified; and (iv) the inactive heterochiral species for bulkier *R*-groups appear to be more thermodynamically stable than the active homochiral ones.

This model has been extensively investigated in experimental kinetic and spectroscopic aspects in combination with theoretical studies. However, other reaction models have been developed, including monomeric-dominant ones, as well as a theoretical model, not supported by experimental results, by Brown and his group (Gridnev *et al.*, 2003). Nevertheless, the organic substrates used are not chemically close to biotic molecules such as amino acids, sugars, or phospholipid precursors, which makes this autocatalytic chiral reaction an uncertain model for prebiotic symmetry breaking.

4. Did Prebiotic Synthesis Only Lead to Racemic Mixtures?

Chemical transformations starting from racemic mixtures of chiral compounds or achiral starting materials give nonchiral systems. The sole possibility involves an initial symmetry breaking and the subsequent amplification of this asymmetry. In other words, something happens to promote this asymmetry from any achiral substrate. If this asymmetry happened first for amino acids or sugars, it is still a subject of debate, and if we consider phospholipids and their precursors, the list of arguments increases exponentially. However, it is certain that once the nonzero enantiomeric excess appeared, amplification played a crucial role in ensuring the presence of enough material. Then, they could

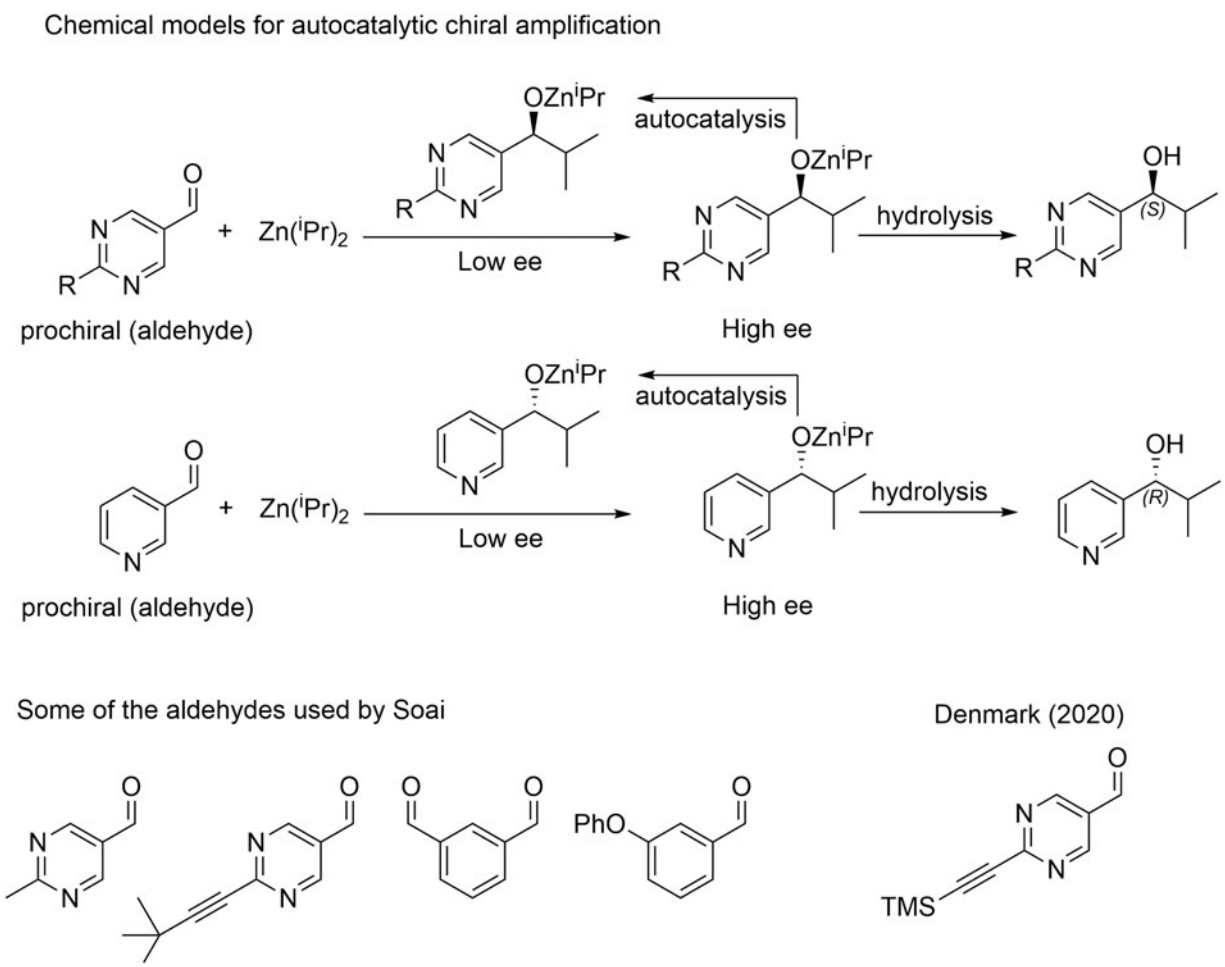

FIG. 4. Examples of autocatalytic chiral amplification and some of the substrates used by Soai *et al.* (2019) and Denmark (Athavale *et al.*, 2020).

start to assemble in biotic, functional polymers such as prebiotic enantiopure peptides, nucleic acids, and, why not, phospholipids.

Two major forces can promote the initial symmetry breaking. The first one is electroweak interactions and the parity violation principle for which D-sugars and L-amino acids are favorably more stable than their opposed enantiomers. The second process that can plausibly explain the symmetry breaking of such molecules is the presence of an achiral environment. For example, a small but relevant number of L-amino acids can be obtained by the degradation of their enantiomers (D-) using CPL. This theory agrees with the ee of L-amino acids found in several meteorites (see the next paragraph). In addition, Earth rotation and/or the presence of chiral crystals (*i.e.*, quartz ones) could be the reason for homochirality in some classes of molecules. Besides, random processes (stochastic) probably played a crucial role in the symmetry breaking of some molecules such as phospholipids (Fiore and Buchet, 2020).

4.1. Prebiotic amino acids synthesis

The first ''prebiotically plausible'' synthesis of α-amino acids (α-AA) is known as the Strecker synthesis. The combination of aldehydes with ammonium chloride in the presence of potassium cyanide (Table 1, Entry 1) forms α-AA as racemic mixture (Benedict, 1929). A significant boost was given by the pioneering experiments of Miller, later revisited by Bada and colleagues (Table 1, Entries 2 and 3) (Bada, 2013). All the experiments carried out can be classified as ''messy'' since the prebiotic conditions set up (bottom-up) are not organized as in a top-down reaction (conventional bench chemistry). Spark discharges, electric discharges (Miller, 1953,1955; Ring *et al.*, 1972; Plankensteiner *et al.*, 2006), volcanic spark discharges (Johnson *et al.*, 2008), or H_2S-enriched discharges (Parker *et al.*, 2011) were used to give energy to a ''soup'' of carbon-containing small molecules in the presence of gases such as CO_2, methane, ammonia, water, and dihydrogen (Airapetian *et al.*, 2016). A large variety of proteogenic and non-proteogenic amino acids were plausibly synthesized as racemic mixtures in those ''prebiotic'' conditions (Table 1, Entries 4–7). The synthetic mechanism is likely to have been of Strecker's type, as further elucidated by Miller himself (Miller, 1957).

Intriguingly, Moran and coworkers showed that a few racemic amino acid mixtures were obtained within protometabolism pathways (Table 1, Entry 8) (Muchowska *et al.*, 2019). Sutherland and colleagues obtained α-AA precursors in so-called ''cyano sulfidic protometabolism'' (Table 1, Entry 9) (Patel *et al.*, 2015). All these molecules were found as racemic mixtures, and apparently, no investigations were carried out to understand whether symmetry breaking occurred when a chiral center was present.

On the contrary, several amino acids were found in Murchinson meteorite extracts (Kvenvolden *et al.*, 1971; Cronin *et al.*, 1981,1985; Engel *et al.*, 1983; Cronin and Pizzarello, 1986; Garcia *et al.*, 2019). In these extracts (Engel and Nagy, 1982; Engel *et al.*, 1990), the predominance of AA with L-configuration was found. Nevertheless, the same predominance was not found in asteroids, as recently reported (Naraoka *et al.*, 2023; Parker *et al.*, 2023).

4.2. Prebiotic synthesis of sugars

Bottom-up experiments, in controlled but untidy conditions, suggest that sugars or phospholipid precursors can be prepared as racemic mixtures under prebiotic conditions such as the *formose* pathway where $Ca(OH)_2$ was used as a catalyzer (Fig. 5) (Butlerow, 1861). Breslow (1959) tentatively rationalized the reaction mechanism: homocoupling of formaldehyde produces glycolaldehyde, which is later converted into glyceraldehyde. Besides, glycolaldehyde, an achiral compound, catalyzes its synthesis. Unfortunately, this compound is not chiral and cannot carry on prebiotic symmetry breaking. From this mixture, a variety of reactions can occur (such as myriad aldol, Cannizzaro, or Lobry de Bruyn–Alberta–van Eckstein ones) (Kim *et al.*, 2011), which lead to a complex set of linear and branched aldo- and ketosugars, all racemic, which contains only 1% of (racemic) ribose mixture. Despite the poor yield, the formose reaction was investigated using some catalyzers such as Pb^{2+} or naturally occurring hydroxyapatite. Eschenmoser and coworkers showed that the variety of products yielded by the formose reaction and the destructive effects that the reaction conditions have on the ribose can be significantly suppressed by the phosphorylation of glycolaldehyde (Müller *et al.*, 1990). Indeed, under alkaline conditions glycoaldehyde phosphate leads to a simple mixture of racemic 2,4 ribose diphosphates. The presence of phosphate groups prevents the Lobry de Bruyn–Alberta–van Eckstein reaction and stabilizes the sugars, providing a plausible route for the synthesis of ribose if the ribose-2-4-diphosphate could be converted to a 5-phosphate or a 1-5-diphosphate (cf. Fig. 4). Recently, the use of Pb^{2+} as a catalyst has increased the conversion of formaldehyde into a mixture of racemic aldopentoses by >30%.

On the contrary, hydroxyapatite enhances the cross-aldol Lobry de Bruyn–Alberda–van Eckstein reaction using the effective positioning of calcium ions on the surface of the minerals (Larralde *et al.*, 1995; Furukawa *et al.*, 2015; Usami and Okamoto, 2017). Thus, tetrose-2-4-diphosphate and hexose-2-4-6-triphosphate are converted into ribose-5-phosphate. In this context, Benner *et al.* (2010, 2012, 2019), investigated whether boranes can form borate complexes, which would stabilize them under these harsh conditions while other sugars would degrade (Kim *et al.*, 2011; Kim and Benner, 2017). Although ribose–borate complexes are more stable than other pentoses, the modest stabilization requires an excess of glycol aldehyde over formaldehyde to inhibit borane and prevent sugar isomer formation with no selectivity for ribose. Modest but real enantioselectivity in glyceraldehyde synthesis was observed when the formose reaction was carried out in the presence of a catalytic number of L-amino acids. However, it was not proven that such reactions were autocatalytic (Blackmond, 2019).

4.3. Prebiotic synthesis of phospholipids

Modern membranes are made of homochiral amphiphilic molecules, such as phospholipids, organized as bilayer membranes (Altamura *et al.*, 2015). The biological selection of homochiral lipids was very likely due to the interesting

Table 1. Synthesis of Amino Acids in Plausible Prebiotic Conditions and Presence of Amino Acids in Meteorites/Asteroids

Entry	*Year*	*Amino acids found*[a]	*Remarks and references*
1	1929	Any α-amino acid	Strecker's synthesis (Benedict, 1929)
2	1953, 1955	Glycine (**Gly**), α-alanine (**Ala**), β-alanine, aspartic acid (**Asp**), α-ABA, sarcosine and two nonidentified compounds	Miller's first experiments (Miller, 1955; Miller, 1953)
3	1972	**Aps**, **Thr**, **Ser**, **Glu**, **Pro**, **Gly**, **Ala**, **Val**, t-Leu, α-NH_2-But, Norvaline, Allo-Ile, **Ile**, **Leu**, Novaleucine, B-Ala, γ-NH_2-But, α-DAP; α-HAB, β-DAB	Analysis of some samples from Miller's experiment (Ring et al., 1972)
4	2008[b]	**Asp**, **Glu**, **Ser**, **Gly**, **Ala**, **Val, Phe**, β-Ala, γ-ABA, β-AIB, β-ABA, α-AIB, α-ABA, iso-valine, norleucine, iso-serine, homo-serine, 2-methyl-serine, β-OH-Asp, iso-valine, norvaline, 2-methyl-glutamic acid, α-AAA, EA, MA, ethanolamine, isopropylamine, *n*-methylamine, and one nonidentified compound.	Analysis of the original samples from Miller's—Bada experiments (Johnson et al., 2008)
5	2011[c]	**Gly**, **Ala**, **Ser**, **Thr**, **Asp**, **Glu**, **Met**, **Val**, **Leu**, **Phe**, β-Ala, iso-serine, homocysteic acid, α-ABA β-ABA, α-AIB, β-AIB, γ-ABI, S-methyl cysteine, iso-valine, methionine sulfoxide, methionine sulfone, iso-leucine, ethionine, MA, EA, cysteamine, ethanolamine	Analysis of the original samples from Miller's—Bada experiments (Parker et al., 2011)

		Aminoacids	*Dipeptides*	*Cyclic dipeptides*	
Experiments carried out in Miller's conditions or simulating plausible prebiotic metabolism					
6	2017 [b,c]	**Gly**, **Ala**, β-**Ala**, **Ser**, α-ABA, β-ABA γ-ABA, **Asp**, **Glu**, **Val**, iso-valine, iso-leucine	Gly-Gly; Gly-Ala; Gly-Thr; Gly-Pro; Pro-Gly; Gly-Val; Val-Gly; Gly-Glu; Glu-Gly; Leu-Gly	(Gly-Gly); (Gly-Pro); (Leu-Gly)	New set of experiments were carried out in the presence of cyanamide as condensing agent (Parker et al., 2014)
7	2006	**Gly, Ala, Val, Ser, Pro, His, Lysine**			Electric dischargers in salty ocean water (Plankensteiner et al., 2006)
8	2019	**Gly**, **Ala**, **Asp** and **Glu**			Product of prebiotic plausible protometabolism (Muchowska et al., 2019)
9	2015	Various amino acid precursors (cf. Fig. 6)			Cyanosulfidic protometabolism (Patel et al., 2015)
Amino acids found in Murchinson meteorite (or asteroids) and their D/L ratio					
10[d]	1982, 1990	**Glu (0.322)**; **Asp (0.202)**; **Pro (0.342)**; **Leu (0.166)**; **Ala (0.682)**			Some amino acids were found in the Murchinson meteorite since 1969 (Cronin and Pizzarello, 1986; Cronin et al., 1985; Cronin et al., 1981; Engel et al., 1983; Garcia et al., 2019; Kvenvolden et al., 1971) with predominancy of L-configuration (Engel and Nagy, 1982; Engel et al., 1990)
11	2023	**Gly, Ala, Val, b-Ala,** α-ABA, β-ABA, **and several isomers of valine**			Some of proteogenic and nonproteogenic amino acids[e] found in asteroid (162873) Ryugu (Naraoka et al., 2023)

[a]Proteogenic amino acids are reported in bold.
[b,c]Amino acids found in the original samples from the three different apparatuses used in Miller's experiments; all amino acids, peptides, and cyclic peptides are found as racemic mixtures.
[d]D/L ratio of some of the 90 amino acids extracted from Murchinson and other meteorites.
[e]Detected as racemic mixtures.
α-AAA = α-amino adipic acid; α-ABA = α-aminobutyric acid; β-AIB = β-aminoisobutyric acid; β-OH-Asp = β-hydroxy aspartic acid; DAB = diaminopropionic acid; DAP = diaminopropionic acid; EA = ethylamine; HAB = hydroxy-γ-aminobutyric; MA = methylamine.

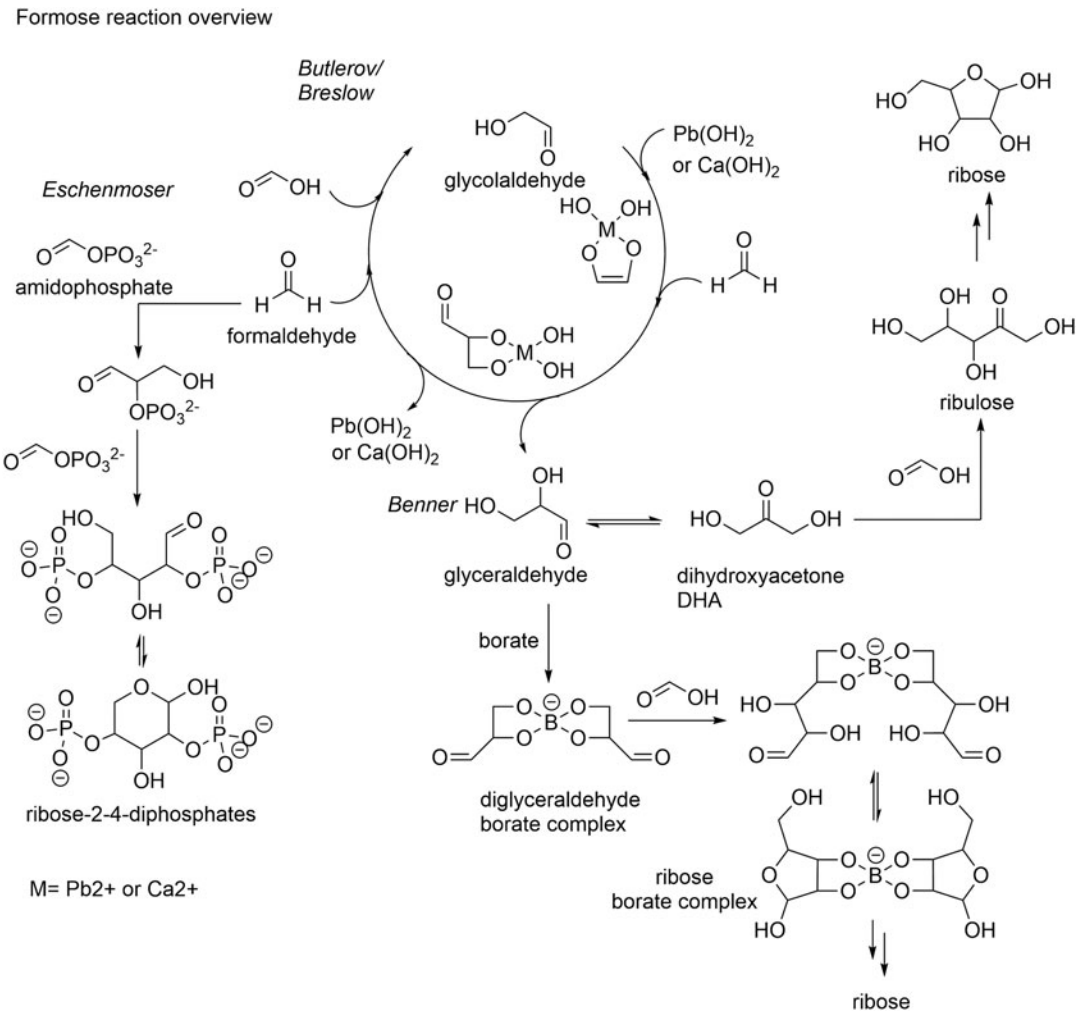

FIG. 5. Schematic summary of the formose reaction and its different modifications by using divalent metal hydroxides or borate complexes (Breslow, 1959; Larralde *et al.*, 1995; Furukawa *et al.*, 2015; Usami and Okamoto, 2017).

properties of the membranes formed. Recently, several authors have explored whether racemic versus scalemic* (Ribó and Hochberg, 2020) versus enantiopure POPC vesicles showed differences in encapsulating a hydrophilic fluorescent dye. Surprisingly, 10 additional percentage of calcein was retained by the enantiopure multilamellar giant vesicles, although calcein is not a chiral compound. Thus, the packaging of enantiopure phospholipids was plausibly more efficient than that of racemic or scalemic (Altamura *et al*., 2020; Fiore and Buchet, 2020). This enhanced potential as a container could be the reason for the selection of homochiral membranes.

According to Wächtershäuser (2003), the Last Universal Common Ancestor probably had homochiral membranes composed of different species of phospholipids such as mixtures of ester and ether ones. However, it is plausible that primitive protocells had membranes made of mixtures of heterochiral amphiphiles (phospholipids or not). This feature would be interesting since heterochiral bilayers are probably more fluid than homochiral bilayers (Eghiaian, 2015; Altamura *et al*., 2020) resulting in higher permeability. Such protocells with enhanced permeability could filter and select more easily possible materials to accumulate and build up primitive polymers from carbohydrates, nucleic bases, amino acids, and their derivatives (Fiore and Buchet, 2020; Mayer, 2022).

Plausible prebiotic precursors of phospholipids are glyceraldehyde, dihydroxyacetone, and glycerol together with their phosphate derivatives such as glycerophosphate (Fig. 6) (Patel *et al*., 2015; Bizzarri *et al*., 2021). The prebiotic preparation of dihydroxyacetone phosphate (Fig. 6) from glyceraldehyde or glycerophosphate, however, remains unexplored (Fiore and Buchet, 2020).

Further investigations have shown that glyceraldehyde, the first chiral product of the ''forms'' reaction (Benner *et al*., 2019), is also present in comets and interstellar ices, opening the perspective of an external source of phospholipid precursors from Earth seeding (Kaiser *et al.*, 2015; Garcia *et al.*, 2019). However, how glycerol was generated or trapped in that interstellar medium remains another unsolved question (Myrgorodska *et al*., 2015). Notably, glyceraldehyde can be phosphorylated in the presence of amido triphosphate followed by pyrophosphate hydrolysis in very good yields, but this reaction leads, again, to racemic mixtures (Krishnamurthy *et al*., 2000).

*Non-racemic.

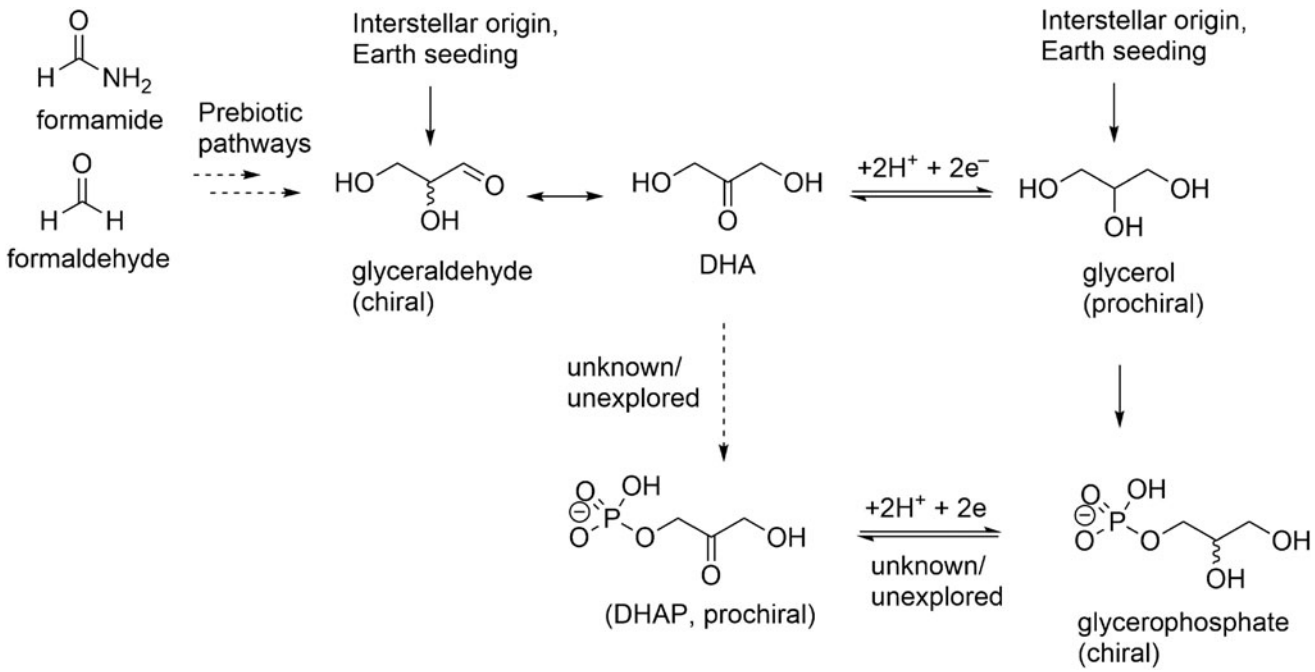

FIG. 6. The interstellar origin of glycerol and glyceraldehyde was independently proven (Kaiser *et al.*, 2015; Garcia *et al.*, 2019). Formamide (Saladino *et al.*, 2012) or formaldehyde (Patel *et al.*, 2015) are considered to be their plausibly prebiotic precursors.

4.4. "One-Pot" Prebiotic Synthesis of Amino Acids, Nucleosides, and Phospholipid Precursors

Recently, Sutherland and colleagues took a major step. They showed that several important classes of biomolecules, such as plausible prebiotic amino acid precursors (Piotto *et al.*, 2019)—2′,3′-cyclic pyrimidine nucleotides and glycerol phosphate—may have hydrogen cyanide (1) and formaldehyde as common chemical precursors in the aforementioned ''cyano sulfidic'' pathway (Patel *et al.*, 2015; Liu *et al.*, 2016; Bonfio *et al.*, 2017, 2019; Sutherland, 2016, 2017; Xu *et al.*, 2017).

The formation of 2′,3′-cyclic pyrimidine nucleotides occurred *via* amino oxazoline intermediate leading to the ''arabinose'' configuration.[†] Remarkably, this is the first example of an enantiopure compound formation from achiral precursors (Fig. 7). However, the concomitant synthesis of the sugar moiety and any purine or pyrimidine nucleobase is the subject of different research.

As shown in Fig. 7, the main issue is the stereo- and regiochemistry at the carbons 2′,3′, and 5′ of the pentose ring. On these bases, any chemist can consider several-step reactions with stereoselection at those carbons. The same group has investigated this direction using multistep reactions that included several stepwise regioselective reactions that would be the key to making a breakthrough (Hein and Blackmond, 2012). The cyanosulfidic mechanism proved a new paradigm as follows: products that are hardly obtained by rational design can instead be successfully produced in a complex environment. Even though, to the best of our knowledge, no studies have been done in this direction, we believe that the cooperation between different sciences, such as experimental and computational chemistry, robotics, and microfluidics technologies, will be a great challenge to explore the transition from simple chemical mixtures to a network of self-evolving systems (Scharf *et al.*, 2015; Cronin, 2017). Certain aspects, however, still require significant and persistent research efforts, which include finding the paths for the emergence of autocatalytic networks of prebiotic relevance (Semenov *et al.*, 2016; Wong and Huck, 2017) or looking for scientific proof for the emergence of biochemical homochirality (Hein and Blackmond, 2012), one of the most important unknowns in the origin of life (OOL) research.

After having demonstrated that a stepwise route from simple abiotic molecules can be used to activate pyrimidine nucleotides (Powner *et al.*, 2009) (Fig. 7), the same authors showed that ribo-, arabinose-, xylo-, and lyxo-configurations can be obtained in a racemic mixture. This was achieved thanks to a multistep reaction *via* the corresponding tetrahydroimidazo[1′,3′]-2″-aminooxazolo-[1′,2′]-pyrimidine (Fig. 8). This molecule is obtained in ''one pot'' from three main components: glyceraldehyde, amino oxazole (obtained from cyanamide and glycolaldehyde, cf. Figs. 7 and 8), and 5-aminoimidazole-4-carboxamide (AICA, cf. Fig. 8).

[†]As reported by the authors (Patel *et al.*, 2015), ''In this synthesis […] the C2 sugar glycolaldehyde (**6**) undergoes phosphate-catalysed condensation with cyanamide (**5** in Fig. 7) to give 2-amino oxazole (**7** in Fig. 10). This heterocycle then participates in a C–C bond-forming reaction with the C3 sugar glyceraldehyde (**11** in Fig.10), which gives rise to a mixture of pentose aminooxazolines. Reaction of the arabino-configured aminooxazoline with cyanoacetylene (**9**) then furnishes the anhydronucleoside, which on heating with phosphate in urea, a by-product of the first step of the sequence, is transformed into ribocytidine-2′,3′-cyclic phosphate (cf. Fig. 10). Ultraviolet irradiation then partially converts this nucleotide into uridine-2′,3′-cyclic phosphate (cf. Fig. 10) and destroys stereoisomeric impurities.''

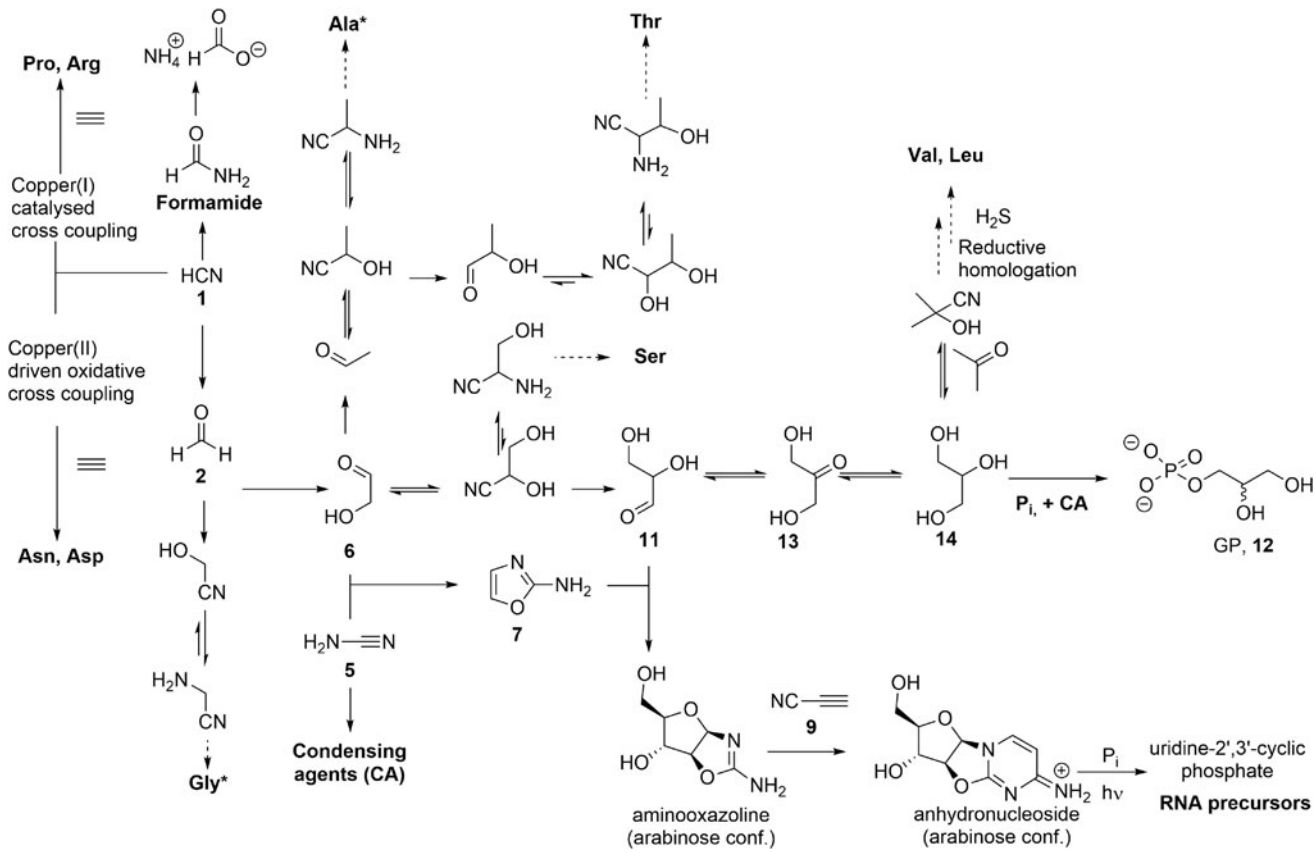

FIG. 7. Plausible routes for the prebiotic synthesis of aminoacids, nucleic acids, and phospholipids. Bottom-up controlled experiments were used to simulate the cyanosulfidic protometabolism; conf., configuration; Adapted from Patel *et al.* (2015).

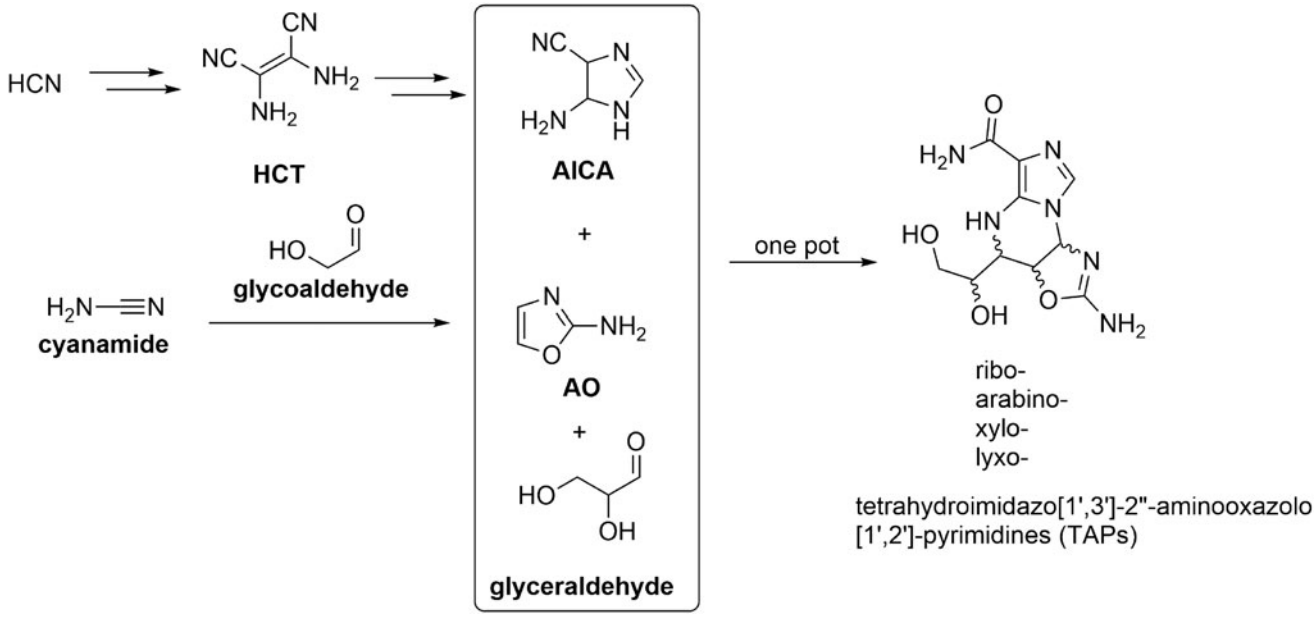

FIG. 8. A key example of multicomponent reaction that allows the one-pot condensation of three plausible prebiotic compounds. Purine precursors or PNA nucleotide precursors can also be obtained from racemic mixtures; adapted from Powner *et al*. (2009, 2010). PNA = peptic nucleic acid.

For instance, AICA was obtained by chemical rearrangement and hydrolysis of hydrogen cyanide tetramers (Powner *et al*., 2010). The tetramerization of HCN (4xHCN) has been known since the 1970s (Orgel *et al*., 1967; Ferris *et al*., 1978; Szabla *et al*., 2014). Different combinations of amino oxazole, AICAs, and glyceraldehyde, or different aldehydes including glycolaldehyde of formaldehyde yielded different potential purine precursors (Powner *et al.*, 2010). Purine precursors or peptide nucleic acid (PNA) (Engelhart and Hud, 2010) precursors were also obtained in racemic mixtures.

5. Symmetry Breaking of Amino Acids, Sugar, and Phospholipid Precursors in Space

5.1. Excess of L-amino acids and D-sugars in meteorites and role of circular polarized light

5.1.1. Amino acids. As mentioned previously, Engel *et al*. (1982) found an excess of some L-amino acids in the neutral and acidified water extracts of the Murchinson meteorite (Table 1, Entry 10). This pioneering research was disputed by Bada, Miller, et al. in a series of *letters* published in ''Nature'' in which both groups advocated their scientific opinions. The former scientists were convinced of their proper way of working, thus no terrestrial contamination, the latter group sustained the theory of terrestrial contamination during the extraction processes (Engel and Nagy, 1983; Engel *et al.*, 1983). A few years later, Engel *et al.* (1990) demonstrated that the AA extracted from the same 1982 samples of the Murchinson meteorite was not contaminated by terrestrial amino acids. The evidence was given by the relevant differences in ^{13}C and ^{15}N isotope abundances between terrestrial and Murchinson samples (Engel *et al*., 1990).

According to these speculations, Meierhenrich *et al.*, among others, showed that the exposure of CPL in simulated interstellar media can induce a mirror symmetry breaking between the formation of L- or D-alanine. Besides, the imbalance depended on the wavelength of the incident CPL and the sense of its rotation (Meinert *et al.*, 2014; Garcia *et al*., 2019). In their opinion, the symmetry imbalance between the two possible stereoisomers of alanine (or other amino acids) occurred before Earth was seeded by asteroids (Naraoka *et al*., 2023; Parker *et al*., 2023) or comet impacts (Myrgorodska *et al.*, 2015), creating the conditions for deracemization before the prebiotic polymerization of peptides and the formation of nucleic acids.

5.1.2. Sugars and sugar precursors. Investigations on the presence of sugars and sugar precursors in Murchison, Murray, and 14 other meteorites showed the presence of complex racemic mixtures of sugars but not a ratio difference for stereoisomers. The presence of racemic mixtures suggested a prebiotic path for this class of molecules without any intervention from CPL or other factors that promote deracemization. However, the methods used for their extraction and identification cannot exclude in this case the external contamination advocated in the case of AA extraction (Garcia *et al*., 2019).

During the last 20 years, thanks to the boost given by Cooper and Rios (2016), the racemic distribution of some short carbon chain sugars was elucidated. Chiral C3 sugar derivative, glyceric acid, was found as racemic mixtures, which is consistent with an abiotic origin (Meinert *et al.*, 2014; Myrgorodska *et al.*, 2015; Garcia *et al.*, 2019). C4 threonic acid is present as a D-enantiomer, with 33–55% of enantiomeric excess (D-ees). Contamination from a terrestrial source was excluded due to the poor amount of these acids in nature. However, C4 sugar alcohols, where threitol was the only chiral one, were present as racemic mixtures in the synthetic sample. Despite their low abundances, the authors found D-ee up to 82% for the C5 sugar acids. The D-ee of arabinonic acid is particularly significant since L-arabinonic acid and L-arabinose are more common than their antipodes. Once again, one can notice that lyxonic acid, a biologically rare sugar acid, also displayed a large D-ee, bolstering the absence of contamination in the experiments. Longer sugars were rarer, though, and their D/L ratios were impossible to determine (Cooper and Rios, 2016). For instance, glyceraldehyde (cf. Figs. 5–8), the first ''chiral'' product of the formose reaction (Benner *et al.*, 2019), was also found in space bodies.

5.2. Symmetry breaking of amino acids driven by chiral crystals

To study the symmetry-breaking phenomenon in racemic amino acid mixtures, enantioselective adsorption experiments were performed on homochiral quartz. Substantial results were reported. The principle of enantioselective adsorption on minerals resembles the principle of liquid chromatography: the mobile phase is composed of a mixture of chemical components passing through the stationary phase that corresponds to a mineral structure. Inorganic quartz presents dextrorotary or levorotatory enantiomorphs, and on primitive Earth, quartz was probably the only enantiomorphic mineral occurring. Therefore, it constitutes the most promising mineral for terrestrial enantioselection. Bonner (1994) experimentally showed the asymmetrical adsorption of alanine by D- or L-quartz, measuring the radioactivity loss when radioactive D- or L-alanine hydrochlorides were adsorbed to the extent of 20 to 30%.

In addition, recent findings about homochiral quartz have confirmed their presence on the early Earth. *S* configuration from D-quartz and *R* from L-quartz were produced, respectively, with a significant ee. In a very recent article, Garcia *et al.* (2021) demonstrated an experimentally significant deracemization (1–55%) of racemic alanine and leucine solutions on homochiral quartz sand.

5.3. Crystallization-induced symmetry breaking: discovering a model for phospholipids

L and D crystals, as separated by Pasteur with his tweezers, are known as ''conglomerates.'' The ''preferential crystallization'' method is well known to ''untangle'' a racemic conglomerate of homochiral *R* and *S* crystals by repeated crystallizations from a supersaturated solution. Kondepudi *et al.* (1990) reported an experimental example of such spontaneous chiral symmetry breaking occurring during the stirred crystallization of a homogeneous solution of achiral sodium chlorate, $NaClO_3$. Symmetry breaking, in this case, appeared in a simple way: if $NaClO_3$ is crystallized from an unstirred solution, statistically equal numbers

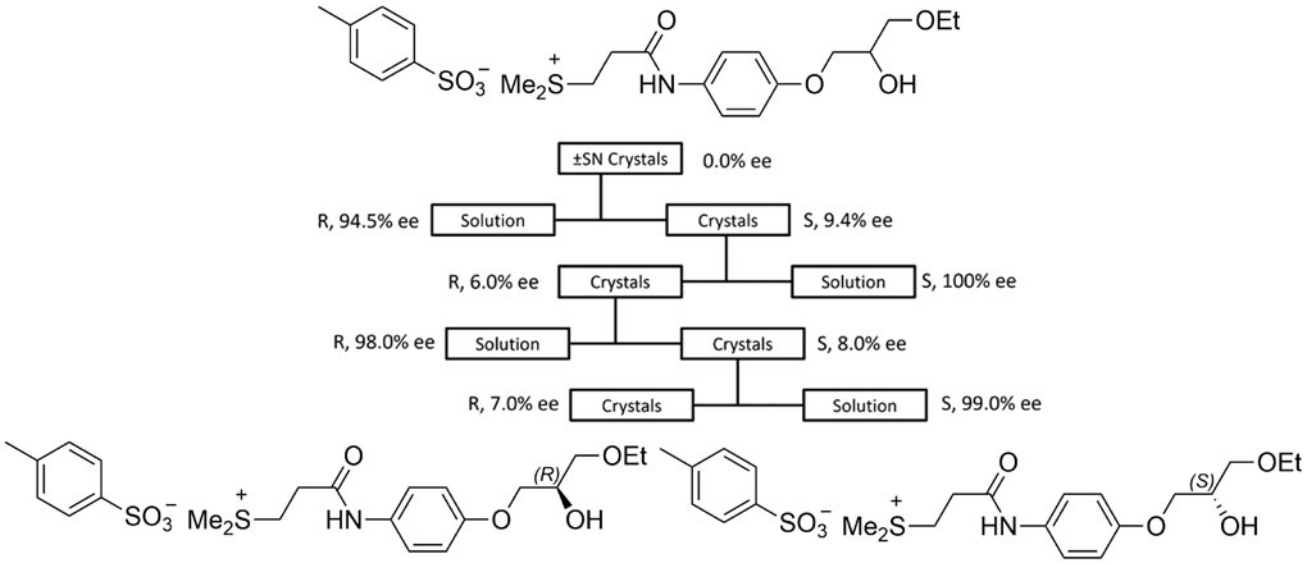

FIG. 9. Enantiomeric resolution by successive recrystallization of (±)-SN in *R* or *S* enantiomers by preferential enrichment. Crystallization conditions are reported as low crystallization in ethanol at r.t. (Tamura *et al.*, 1998). r.t., room temperature.

of L and D crystals[‡] are obtained. However, if the crystallization occurs during rapid stirring, >99% of the crystals are obtained with the same chirality (stochastically D or L) (Kondepudi *et al.*, 1990).

Some years later, Viedma (2004) described in their studies a series of experiments to explore this phenomenon further. They reported that, in primary and secondary nucleation, crystals formed from the solution itself. Nevertheless, secondary nucleation is the process by which a ''mother crystal'' generates secondary crystals at a fast rate, if the solution is stirred. In the case of $NaClO_3$, a first D or L ''mother crystal'' forms new crystals with the corresponding chirality in a few minutes. It is quite evident that the complete chiral purity reported in Viedma's experiments could not be explained only by the model of an initial single chiral crystal. Considering the role of grinding glass balls in the symmetry-breaking phenomenon, Viedma suggested that under these conditions a continuous dissolution–crystallization process would be highly enhanced. Thus, the system would be very similar to a homogeneous autocatalytic reaction. These results can provide a more prebiotic view, indicating how essential chiral molecules would be formed through conglomerate solids (Kondepudi *et al.*, 1990; Viedma, 2004; Gridnev, 2006).

[‡]L and D are currently used to differentiate two different crystals that show asymmetry in their crystalline structure even though they are made from achiral compounds as in the case of $NaClO_3$; to clarify this, we cite from Viedma's article the following sentences: [''the most important factor in this chiral symmetry breaking is secondary nucleation. In primary nucleation crystals are generated from the solution itself. On the other hand, secondary nucleation is the process by which a 'mother crystal' generates secondary crystals at a fast rate if the solution is stirred; in the case of $NaClO_3$, a first L or D 'mother crystal' generates new crystals with similar chirality in a few minutes. The sequence of events is as follows: a single crystal randomly nucleates at some time by primary nucleation and begins to grow. When the crystal reaches a critical size, it begins to produce secondary nuclei, which have the same chirality as their 'mother'''].

Moreover, Tamura *et al.* (1998, 2010) proposed a phenomenon referred to as ''preferential enrichment,'' which is completely opposed to preferential crystallization. In preferential enrichment it is the mother liquor that is enantiomerically enriched by recrystallization, whereas the crystals result in scalemic mixtures of enantiomers as summarized in Fig. 9 (Tamura *et al.*, 1998, 2010). Besides, the molecules prepared by Tamura *et al.* (1998, 2010) are close to lysophospholipids in archaea and could be favorable substrates with which to study homochiral symmetry breaking in phospholipids by using similar structures to model molecules (Skorda *et al.*, 2022).

6. Conclusions

In this review, we have focused our attention on some aspects of the OOL studies that are poorly tackled: homochirality emergence in prebiotic environments and simulated prebiotic ones. The prebiotic Earth was likely composed of messy environments characterized by ''one-pot'' conditions where numerous compounds would be present at once. In this review, we have highlighted that symmetry breaking probably occurred due to different factors, and two main aspects can be pointed out as crucial. The first is the seeding from meteorites or other space bodies. One clear example is the seeding of scalemic (nonracemic) amino acid mixtures that broke the equilibria on an early abiotic (and plausibly racemic) Earth. This has not yet been proven for sugars or phospholipid precursors.

Furthermore, symmetry breaking of racemic mixtures on Earth cannot be excluded *a priori*. Symmetric autocatalysis and/or preferential enrichment processes, driven by the alternation of wet/dry (day and night) cycles for instance, might have favored the casual accumulation of one enantiomer over the other. Experimentally, the systems chemistry approach at the laboratory scale is probably one of the most reliable ways to understand life's emergence and the homochiral symmetry breaking of biotic molecules with

the use of aqueous diluted mixtures of reactants. This was the direction taken by Blackmond *et al.* (2001), Hein *et al.* (2011), Hein and Blackmond (2012), and Blackmond (2020), among several scientists, with the reproduction of some experiments at the laboratory scale but not properly in messy (read prebiotic/uncontrolled/bottom-up) conditions. Although the results were astonishing, the conditions used were still controlled, and no other examples of this kind of chemistry have been reported.

One of the leads for symmetry breaking could be the presence of a chiral inductor in a protometabolism pathway. Indeed, the presence of a scalemic amount of any plausible chiral inducer (such as L-proline) can lead to high enantiomeric resolution. However, adding a chiral inductor to a protometabolism path can be one of the answers to the chicken-egg *dilemma*: the presence of a scalemic amount of any plausible chiral inducer (cf. L-proline) can lead to a high enantiomeric resolution as summarized in one of her reviews (Blackmond, 2010).

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Abbreviations Used

AICA = 5-aminoimidazole-4-carboxamide
ATP = adenosine triphosphate
CPL = circularly polarized light
ee = enantiomeric enrichment
PNA = peptic nucleic acid
OOL = origin of life